2026/2027 (Physics) Teaching Plan

Teaching Level: Form 4

Curriculum (if any; e.g. IGCSE / IAL):

- **IGCSE**

Textbook:

1. **Pearson Edexcel International GCSE (9-1) Physics Student Book**

Marks Allocation

1st Term (Formative)	**Assessment**	**Homework**	**Classwork**	**Participation**	**Experiment**	**Test**	**Quiz**	**Total**
	No. of Times	**4-6**	**6-8**	**--**	**1**	**2**	**1**	**--**
	%	**15**	**20**	**10**	**10**	**35**	**10**	**100%**
	Supplementary Assessment [1]	**N**	**N**	**N**	**N**	**Y**	**Y**	**--**
2nd Term (Formative)	**Assessment**	**Homework**	**Classwork**	**Participation**	**Experiment**	**Test**	**Quiz**	**Total**
	次數	**4-6**	**6-8**	**--**	**1**	**2**	**1**	**--**
	%	**15**	**20**	**10**	**10**	**35**	**10**	**100%**
	Supplementary Assessment [1]	**N**	**N**	**N**	**N**	**Y**	**Y**	**--**

[1] 必須補測/交項目，每次完成評核都須登記入紅手冊系統以通知家長

Week	Unit	Contents	Teaching Objectives	BAA (code)	Evaluation Method
1-10	Force and Motion	Motion	Students are expected to: • be able to use the equations for uniformly accelerated motion in one dimension: $s=\frac{(u+v)t}{2}$ $v=u+at$ $s=ut+\frac{1}{2}at^2$ $v^2=u^2+2as$	A-5 A-6 A-7 A-8 A-9 A-10	Homework Classwork Experiment Week 6: Quiz 1 Week 10: Test 1
		Force resolution	Students are expected to: • be able to resolve a vector into two components at right angles to each other by drawing and by calculation • be able to find the resultant of two coplanar vectors at any angle to each other by drawing, and at right angles to each other by calculation • be able to draw and interpret free-body force diagrams to represent forces on a particle or on an extended but rigid body using the concept of centre of gravity of an extended body • be able to use the equation $\sum F=ma$, and understand how to use this equation in situations where m is constant (Newton's second law of motion), including Newton's first law of motion where a = 0, objects at rest or travelling at constant velocity • Use of the term 'terminal velocity' is expected.		

		Chapter 4: Momentum	Student are expected to: • know and use the relationship between momentum, mass and velocity • use the idea of momentum to explain safety features • use the conservation of momentum to calculate the mass, velocity or momentum of objects • use the relationship between force, change in momentum and time taken • understand that momentum is defined as $p = mv$ • know the principle of conservation of linear momentum, understand how to relate this to Newton's laws of motion and understand how to apply this to problems in one dimension • understand how to use the equation $impulse = F\Delta t = \Delta p$ (Newton's second law of motion) • understand how to apply conservation of linear momentum to problems in two dimensions • understand how to determine whether a collision is elastic or inelastic • be able to derive and use the equation $E_k = \frac{p^2}{2m}$ for the kinetic energy of a non-relativistic particle		

		Chapter 5: The turning effect of forces	Student are expected to: • know and use the relationship between the moment of a force and its perpendicular distance from the pivot • know that the weight of a body acts through its centre of gravity • use the principle of moments for a simple system of parallel forces acting in one plane • understand how the upward forces on a light beam, supported at its ends, vary with the position of a heavy object placed on the beam • be able to use the equation for the moment of a force, $moment\ \ of\ \ force = Fx$ where x is the perpendicular distance between the line of action of the force and the axis of rotation • be able to use the concept of centre of gravity of an extended body and apply the principle of moments to an extended body in equilibrium		

11-17	Energy Resources and Energy Transfer	Chapter 14: Energy transfer	Students are expected to: • describe energy transfers involving energy stores: energy stores: chemical, kinetic, gravitational, elastic, thermal, magnetic, electrostatic, nuclear energy transfers: mechanically, electrically, by heating, by radiation (light and sound) • use the principle of conservation of energy • know and use the relationship between efficiency, useful energy output and total energy output: $efficiency = \frac{useful\ \ energy}{output\ \ efficiency} \times 100\,\%$ • describe a variety of everyday and scientific devices and situations, explaining the transfer of the input energy in terms of the above relationship, including their representation by Sankey diagrams • explain ways of reducing unwanted energy transfer, such as insulation	A-1 A-2 A-5 A-6 A-7 A-8 A-9 A-10 B-1 B-2 D-6	Week 15: Test 2

		Chapter 16: Work and power	• use the following units: kilogram (kg), joule (J), metre (m), metre/second (m/s), metre/second^2 (m/s^2), newton (N), second (s) and watt (W) • know and use the relationship between work done, force and distance moved in the direction of the force: work done = force × distance moved W = F × d • know that work done is equal to energy transferred • know and use the relationship between gravitational potential energy, mass, gravitational field strength and height: gravitational potential energy = mass × gravitational field strength × height GPE = m × g × h • know and use the relationship: $kinetic\ \ energy = \frac{1}{2} \times mass \times spee\, d^2$ $KE = \frac{1}{2} \times m \times v^2$ • understand how conservation of energy produces a link between gravitational potential energy, kinetic energy and work • describe power as the rate of transfer of energy or the rate of doing work • use the relationship between power, work done (energy transferred) and time taken: $power\ = \frac{work\ \ done}{time\ \ taken}$ • $P = \frac{W}{t}$		

		Chapter 17: Energy resources and electricity generation	• describe the energy transfers involved in generating electricity using wind, water, geothermal resources, solar heating systems, solar cells, fossil fuels and nuclear power • describe the advantages and disadvantages of methods of large-scale electricity production from various renewable and non-renewable resources		
17-18	Christmas Holiday				
19	Study week				
20	Exam				

21-26	Waves	Chapter 10: Properties of waves	Students are expected to: • use the following units: degree (°), hertz (Hz), metre (m), metre/second (m/s) and second (s) • explain the difference between longitudinal and transverse waves • know the definitions of amplitude, wavefront, frequency, wavelength and period of a wave • know that waves transfer energy and information without transferring matter • know and use the relationship between the speed, frequency and wavelength of a wave: $wave\ \ speed\ = frequency\ \times\ wavelength$ $v\ =\ f\ \times\ \lambda$ • use the relationship between frequency and time period: $frequency = \frac{1}{time\ \ period}$ $f = \frac{1}{T}$ • use the above relationships in different contexts, including sound waves and electromagnetic waves • explain why there is a change in the observed frequency and wavelength of a wave when its source is moving relative to an observer and that this is known as the Doppler effect • explain that all waves can be reflected and refracted	A-1 A-2 B-1 B-8 C-3 D-3 D-4 D-7	Homework Classwork Week X: Test 1

		Chapter 11: The electromagnetic spectrum	• know that light is part of a continuous electromagnetic spectrum that includes radio, microwave, infrared, visible, ultraviolet, x-ray and gamma ray radiations, and that all these waves travel at the same speed in free space • know the order of the electromagnetic spectrum in terms of decreasing wavelength and increasing frequency, including the colours of the visible spectrum • explain some of the uses of electromagnetic radiations, including: radio waves: broadcasting and communications microwaves: cooking and satellite transmissions infrared: heaters and night vision equipment visible light: optical fibres and photography ultraviolet: fluorescent lamps x-rays: observing the internal structure of objects and materials, including for medical applications gamma rays: sterilising food and medical • explain the detrimental effects of excessive exposure of the human body to electromagnetic waves, including: microwaves: internal heating of body tissue infrared: skin burns ultraviolet: damage to surface cells and blindness gamma rays: cancer, mutation and describe simple protective • measures against the risks		

		Chapter 12: Light waves	• know that light waves are transverse waves and that they can be reflected and refracted • use the law of reflection (the angle of incidence equals the angle of reflection) • draw ray diagrams to illustrate reflection and refraction • investigate the refraction of light, using rectangular blocks, semi-circular blocks and triangular prisms (practical) • know and use the relationship between refractive index, angle of incidence and angle of refraction: $n = \frac{\sin i}{\sin r}$ • investigate the refractive index of glass, using a glass block (practical) • describe the role of total internal reflection in transmitting information along optical fibres and in prisms • explain the meaning of critical angle c • know and use the relationship between critical angle and refractive index: $\sin c = \frac{1}{n}$		

		Chapter 13: Sound	• know that sound waves are longitudinal waves that can be reflected and refracted • know that the frequency range for human hearing is 20–20 000 Hz • investigate the speed of sound in air • understand how an oscilloscope and microphone can be used to display a sound wave • investigate the frequency of a sound wave using an oscilloscope • understand how the pitch of a sound relates to the frequency of vibration of the source • understand how the loudness of a sound relates to the amplitude of vibration of the source		
23-24	Lunar New Year Holiday				

27-31	Solids, Liquids and Gases	Chapter 18: density and pressure	Students are expected to: • use the following units: degree Celsius (°C), Kelvin (K), joule (J), kilogram (kg), kilogram/metre3 (kg/m^3), metre (m), $metre^2$ (m^2), $metre^3$ (m^3), metre/second (m/s), $metre/second^2$ (m/s^2), newton (N) and pascal (Pa) • use the following unit: joules/kilogram degree Celsius (J/kg °C) • know and use the relationship between density, mass and volume: $density = \frac{mass}{volume}$ $\rho = \frac{m}{V}$ • investigate density using direct measurements of mass and volume (practical) • know and use the relationship between pressure, force and area: $pressure = \frac{force}{area}$ $p = \frac{F}{A}$ • understand how the pressure at a point in a gas or liquid at rest acts equally in all directions • know and use the relationship for pressure difference: *pressure difference=height×density×gravitational field strength* • $p = h \times \rho \times g$	A-1 A-2 A-4 A-7 A-8 A-9 B-1 B-3	Homework Classwork

31-36	Energy Resources and Energy Transfer	Chapter 15: Thermal energy	• describe how thermal energy transfer may take place by conduction, convection and radiation • explain the role of convection in everyday phenomena • explain how emission and absorption of radiation are related to surface and temperature • investigate thermal energy transfer by conduction, convection and radiation (practical)	A-1 A-2 A-5 A-6 A-7 A-8 A-9 A-10 B-1 B-2 D-6	
36-42	Solids, Liquids and Gases	Chapter 19: Solids, liquids and gases	• explain why heating a system will change the energy stored within the system and raise its temperature or produce changes of state • describe the changes that occur when a solid melts to form a liquid, and when a liquid evaporates or boils to form a gas • describe the arrangement and motion of particles in solids, liquids and gases • obtain a temperature–time graph to show the constant temperature during a change of state (practical) • know that specific heat capacity is the energy required to change the temperature of an object by one degree Celsius per kilogram of mass (J/kg °C) • use the equation: *change in thermal energy=mass×specific heat capacity×change in temperature* $\Delta Q = m \times c \times \Delta T$ • investigate the specific heat capacity of materials including water and some solids	A-1 A-2 A-4 A-7 A-8 A-9 B-1 B-3	Homework Classwork (Experiment) Week X: Test 2

			• explain how molecules in a gas have random motion and that they exert a force, and hence a pressure, on the walls of a container • understand why there is an absolute zero of temperature, which is –273 °C • describe the Kelvin scale of temperature and be able to convert between the Kelvin and Celsius scales • understand why an increase in temperature results in an increase in the average speed of gas molecules • know that the Kelvin temperature of a gas is proportional to the average kinetic energy of its molecules • explain, for a fixed amount of gas, the qualitative relationship between: pressure and volume at constant temperature pressure and Kelvin temperature at constant volume • use the relationship between the pressure and Kelvin temperature of a fixed mass of gas at constant volume: $\frac{p_1}{T_1} = \frac{p_2}{T_2}$ • use the relationship between the pressure and volume of a fixed mass of gas at constant temperature: $p_1 V_1 = p_2 V_2$ • be able to use the equations $\Delta E = mc\Delta\theta$ and $\Delta E = L\Delta m$ • understand the concept of internal energy as the random distribution of potential and kinetic energy amongst molecules • understand the concept of absolute zero and how the average kinetic energy of molecules is related to the absolute temperature • be able to use the equation $pV = NkT$ for an ideal gas		

			• be able to derive and use the equation $\frac{1}{2}m<c^2\geq\frac{3}{2}kT$		
43	Study week				
44	Exam				